Samkedemia
EDUCATION

ULTIMATE SOLUTION BOOK:
ALL EXERCISES & PAST EXAMS

Mathematics for Business A

KU LEUVEN • BACHELOR OF BUSINESS ADMINISTRATION

VERSION: 10 NOV 2026

LET ME INTRODUCE MYSELF.

About Me

I'll make sure you pass this exam. I know exactly what it takes to get there.



BACKGROUND

- My name is **Jovan Samardzic**, I am 26 years old
- Got my **MSc in Statistics at KU Leuven**, graduated **Magna Cum Laude**
- Recipient of the **Science@Leuven Scholarship**

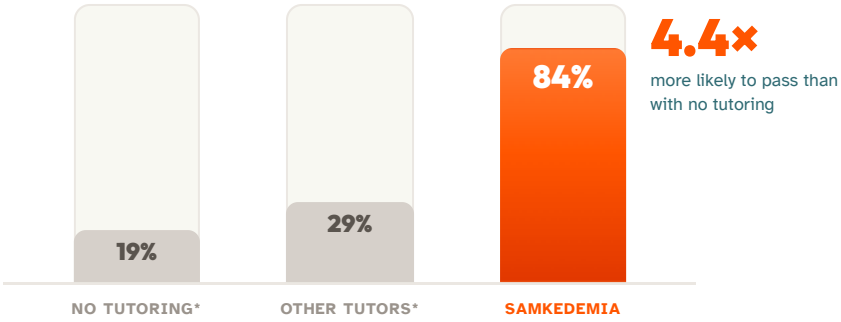
STYLE

- Founder of **Samkedemia**, helping students pass their exams
- Worked with **200+ university students** from all kinds of backgrounds
- **Plain language, not textbook theory** - just what actually makes sense

PROOF THAT IT WORKS.

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* BASED ON INTERNAL RESEARCH CONDUCTED IN 2026.

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Chapter 1 - Exponential and Logarithmic Functions

Question 1

Let x stand for any strictly positive number. Write the following expressions as a single power of x.

(a) $\left(x^{0.5}\right)^3$

(b) $x^2 \sqrt[3]{x}$

(c) $\left(\frac{1}{x^{0.25}}\right)^6$

(d) $\frac{x^3\sqrt{x}}{\sqrt[4]{x^3}}$

Tip: Exponent and Root Rules

The following rules let you rewrite any combination of powers and roots as a single power.

Power of a Power Rule: $(b^x)^y = b^{x \cdot y}$

Power of a Product Rule: $(a \cdot b)^x = a^x \cdot b^x$

Power of a Fraction Rule: $\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$

Product Rule: $b^x \cdot b^y = b^{x+y}$

Quotient Rule: $\frac{b^x}{b^y} = b^{x-y}$

Negative Exponent Rule: $b^{-x} = \frac{1}{b^x}$

Roots as Fractional Exponents: $\sqrt[n]{b} = b^{\frac{1}{n}}$

Solution 1: (a) $\left(x^{0.5}\right)^3$

Apply the power of a power rule:

$$\left(x^{0.5}\right)^3 = x^{0.5 \cdot 3} = x^{1.5} = x^{\frac{3}{2}}$$

Answer: The expression simplifies to $x^{\frac{3}{2}}$.

Solution 1: (b) $x^2 \sqrt[3]{x}$

Rewrite the cube root as a fractional exponent, then apply the product rule:

$$x^2 \sqrt[3]{x} = x^2 \cdot x^{\frac{1}{3}} = x^{2+\frac{1}{3}} = x^{\frac{7}{3}}$$

Answer: The expression simplifies to $x^{\frac{7}{3}}$.

Solution 1: (c) $\left(\frac{1}{x^{0.25}}\right)^6$

Rewrite the fraction using the negative exponent rule, then apply the power of a power rule:

$$\left(\frac{1}{x^{0.25}}\right)^6 = \left(x^{-0.25}\right)^6 = x^{-0.25 \cdot 6} = x^{-1.5} = x^{-\frac{3}{2}}$$

Answer: The expression simplifies to $x^{-\frac{3}{2}}$.

Solution 1: (d) $\dfrac{x^3\sqrt{x}}{\sqrt[4]{x^3}}$

Rewrite both roots as fractional exponents, then apply the product and quotient rules:

$$\frac{x^3\sqrt{x}}{\sqrt[4]{x^3}} = \frac{x^3 \cdot x^{\frac{1}{2}}}{x^{\frac{3}{4}}} = x^{3+\frac{1}{2}-\frac{3}{4}} = x^{\frac{11}{4}}$$

Answer: The expression simplifies to $x^{\frac{11}{4}}$.

Question 2

Simplify the following expressions.

(a) $\left(x^2y^3\right)^4$

(b) $x^{\frac{2}{3}}y^{\frac{1}{2}}x^{\frac{3}{4}}y^{\frac{1}{3}}$

(c) $\left(\frac{x}{yz}\right)^2\left(\frac{y}{xz}\right)^2\left(\frac{z}{xy}\right)^2$

Tip: Exponent and Root Rules

The following rules let you rewrite any combination of powers and roots as a single power.

Power of a Power Rule: $(b^x)^y = b^{x\cdot y}$

Power of a Product Rule: $(a\cdot b)^x = a^x\cdot b^x$

Power of a Fraction Rule: $\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$

Product Rule: $b^x\cdot b^y = b^{x+y}$

Quotient Rule: $\frac{b^x}{b^y} = b^{x-y}$

Negative Exponent Rule: $b^{-x} = \frac{1}{b^x}$

Roots as Fractional Exponents: $\sqrt[n]{b} = b^{\frac{1}{n}}$

Solution 2: (a) $\left(x^2y^3\right)^4$

Apply the power of a product rule:

$$\left(x^2y^3\right)^4 = x^{2\cdot4}y^{3\cdot4} = x^8y^{12}$$

Answer: The expression simplifies to x^8y^{12}.

Solution 2: (b) $x^{\frac{2}{3}}y^{\frac{1}{2}}x^{\frac{3}{4}}y^{\frac{1}{3}}$

Group the terms with the same base, then apply the product rule to each base separately:

$$\begin{aligned} x^{\frac{2}{3}}y^{\frac{1}{2}}x^{\frac{3}{4}}y^{\frac{1}{3}} &= \left(x^{\frac{2}{3}}x^{\frac{3}{4}}\right)\left(y^{\frac{1}{2}}y^{\frac{1}{3}}\right) \\ &= x^{\frac{2}{3}+\frac{3}{4}}y^{\frac{1}{2}+\frac{1}{3}} \\ &= x^{\frac{8}{12}+\frac{9}{12}}y^{\frac{3}{6}+\frac{2}{6}} = x^{\frac{17}{12}}y^{\frac{5}{6}} \end{aligned}$$

Answer: The expression simplifies to $x^{\frac{17}{12}}y^{\frac{5}{6}}$.

Solution 2: (c) $\left(\frac{x}{yz}\right)^2\left(\frac{y}{xz}\right)^2\left(\frac{z}{xy}\right)^2$

Apply the power of a fraction rule to each factor, then combine the fractions and apply the

quotient rule:

$$\left(\frac{x}{yz}\right)^2\left(\frac{y}{xz}\right)^2\left(\frac{z}{xy}\right)^2 = \frac{x^2}{y^2z^2}\cdot\frac{y^2}{x^2z^2}\cdot\frac{z^2}{x^2y^2}$$

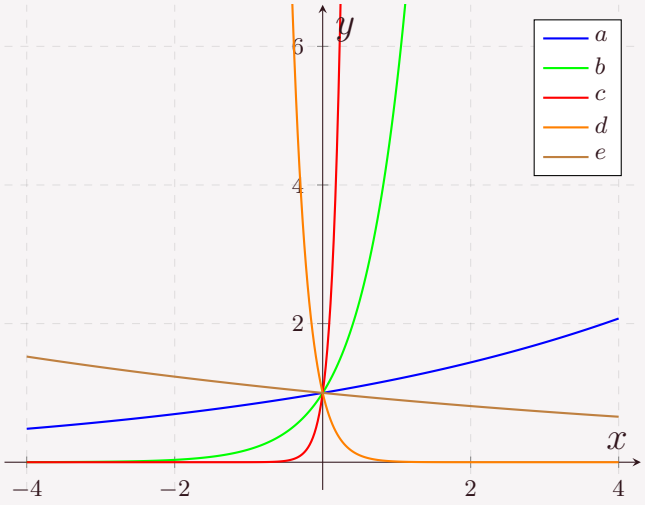
$$= \frac{x^2y^2z^2}{x^4y^4z^4}$$

$$= x^{2-4}y^{2-4}z^{2-4} = x^{-2}y^{-2}z^{-2}$$

Answer: The expression simplifies to $x^{-2}y^{-2}z^{-2}$.

Question 3

The graphs of five exponential functions are drawn in the following figure. The bases that were used are 2500, 0.01, 1.2, $2e$, and 0.9. Find out which graph corresponds to which exponential function.



Tip: Identifying Graphs of Exponential Functions

Exponential functions $y = a^x$ behave differently based on their base a:

- If $a > 1$, the graph increases as x increases. The larger the base, the faster it grows.
- If $0 < a < 1$, the graph decreases as x increases. The closer the base is to 0, the faster it decays.

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Solution 3: Match each graph to the base of its exponential function

Compare each curve's growth or decay speed to the five given bases, from the most extreme curves to the calmest ones.

Step 1: Find the fastest-growing curve.

Curve c shoots up almost vertically right at $x = 0$, reaching $y = 6$ before x even leaves 0. Since a larger base grows faster, this curve must belong to the largest base given, 2500.

Step 2: Find the fastest-decaying curve.

Curve d is the mirror image of curve c: it drops almost vertically to 0 just to the left of $x = 0$, then stays flat for $x > 0$. Since a base closer to 0 decays faster, this curve must belong to the base closest to 0, namely 0.01.

Step 3: Find the moderately fast-growing curve.

Curve b also grows as x increases, but far more gently than c - it only reaches $y = 6$ around $x \approx 1$. Of the two remaining bases above 1 (1.2 and $2e$), this faster growth matches the larger one, $2e \approx 5.4366$.

Step 4: Match the two remaining, slowly-changing curves.

Curve a increases only slowly across the whole picture, which matches the one remaining base above 1, namely 1.2. Curve e decreases just as slowly in the opposite direction, which matches the one remaining base below 1, namely 0.9.

Answer:

$a \rightarrow$ base 1.2

$b \rightarrow$ base $2e$

$c \rightarrow$ base 2500

$d \rightarrow$ base 0.01

$e \rightarrow$ base 0.9

Question 4

A function f has an equation of the form:

$$y = A \cdot b^x,$$

where A and b are real numbers. The graph of f contains the points $(0.5, 1)$ and $(2.5, 9)$.

Determine the equation of f, i.e., find the values of A and b.

Tip: Solving for Two Unknowns by Dividing Two Equations

When two equations share the same two unknowns and each is a constant times a power with the same base, dividing one by the other cancels the common constant and leaves a single equation in one unknown:

$$\frac{c \cdot k^{t_1}}{c \cdot k^{t_2}} = k^{t_1 - t_2}$$

Solution 4: Find the equation of f

Step 1: Write one equation per given point.

Substitute each point into $y = A \cdot b^x$:

$$1 = A \cdot b^{0.5}$$

$$9 = A \cdot b^{2.5}$$

Step 2: Divide the two equations to eliminate A.

$$\frac{9}{1} = \frac{A \cdot b^{2.5}}{A \cdot b^{0.5}} = b^2$$

Step 3: Solve for b, then back-substitute for A.

Take the square root of both sides:

$$b = (9)^{1/2} = 3$$

Substitute $b = 3$ into the first equation:

$$1 = A \cdot 3^{0.5} \quad \Rightarrow \quad A = \frac{1}{\sqrt{3}}$$

Step 4: Write the final equation of f.

$$f(x) = \frac{1}{\sqrt{3}} \cdot 3^x$$

Answer: $A = \frac{1}{\sqrt{3}}$ and $b = 3$, so $f(x) = \frac{1}{\sqrt{3}} \cdot 3^x$.

Question 5

If possible, calculate the following without the use of a calculator.

(a) $\log_2 1$

(b) $\log_2\left(\frac{1}{8}\right)$

(c) $\log_2\left(\sqrt[4]{2^5}\right)$

(d) $\log_2\left(\frac{1}{\sqrt[3]{2}}\right)$

(e) $\log_{\frac{1}{3}} 9$

(f) $\log_3 10$

(g) $\log(1\,000\,000)$

(h) $\log(10^{12})$

(i) $\log(0.001)$

(j) $\ln 1$

(k) $\ln\sqrt{e}$

(l) $\ln\left(\frac{1}{e}\right)$

Tip: Logarithm Rules

The following rules apply to logarithms in any base b.

Product Rule: $\log_b(x \cdot y) = \log_b(x) + \log_b(y)$

Quotient Rule: $\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$

Power Rule: $\log_b(x^r) = r \cdot \log_b(x)$

Change of Base: $\log_b(x) = \frac{\ln(x)}{\ln(b)}$

Logarithm of the Base: $\log_b(b) = 1$

Logarithm of One: $\log_b(1) = 0$

Common Logarithm: $\log(x) = \log_{10}(x)$

Natural Logarithm: $\ln(x) = \log_e(x)$

Solution 5: (a) $\log_2 1$

Apply the logarithm of one rule:

$$\log_2 1 = 0$$

Answer: The expression evaluates to 0.

Solution 5: (b) $\log_2\left(\frac{1}{8}\right)$

Rewrite $\frac{1}{8}$ as a power of 2, then apply the power rule:

$$\log_2\left(\frac{1}{8}\right) = \log_2\left(2^{-3}\right) = -3 \cdot \log_2(2) = -3 \cdot 1 = -3$$

Answer: The expression evaluates to -3.

Solution 5: (c) $\log_2\left(\sqrt[4]{2^5}\right)$

Rewrite the fourth root as a fractional exponent, then apply the power rule:

$$\log_2\left(\sqrt[4]{2^5}\right) = \log_2\left(2^{\frac{5}{4}}\right) = \frac{5}{4} \cdot \log_2(2) = \frac{5}{4} \cdot 1 = \frac{5}{4}$$

Answer: The expression evaluates to $\frac{5}{4}$.

Solution 5: (d) $\log_2\left(\frac{1}{\sqrt[3]{2}}\right)$

Rewrite the cube root and the fraction as a single power of 2, then apply the power rule:

$$\log_2\left(\frac{1}{\sqrt[3]{2}}\right) = \log_2\left(2^{-\frac{1}{3}}\right) = -\frac{1}{3}\cdot\log_2(2) = -\frac{1}{3}\cdot 1 = -\frac{1}{3}$$

Answer: The expression evaluates to $-\frac{1}{3}$.

Solution 5: (e) $\log_{\frac{1}{3}} 9$

Rewrite 9 as a power of $\frac{1}{3}$, then apply the power rule:

$$\log_{\frac{1}{3}}(9) = \log_{\frac{1}{3}}\left(\left(\frac{1}{3}\right)^{-2}\right) = -2\cdot\log_{\frac{1}{3}}\left(\frac{1}{3}\right) = -2\cdot 1 = -2$$

Answer: The expression evaluates to -2.

Solution 5: (f) $\log_3 10$

Since 10 cannot be written as a power of 3, this logarithm has no exact value that can be found without a calculator.

Answer: This cannot be computed exactly without a calculator.

Solution 5: (g) $\log(1\,000\,000)$

Rewrite $1\,000\,000$ as a power of 10, then apply the power rule:

$$\log(1\,000\,000) = \log(10^6) = 6 \cdot \log(10) = 6 \cdot 1 = 6$$

Answer: The expression evaluates to 6.

Solution 5: (h) $\log(10^{12})$

Apply the power rule directly:

$$\log(10^{12}) = 12 \cdot \log(10) = 12 \cdot 1 = 12$$

Answer: The expression evaluates to 12.

Solution 5: (i) $\log(0.001)$

Rewrite 0.001 as a power of 10, then apply the power rule:

$$\log(0.001) = \log(10^{-3}) = -3 \cdot \log(10) = -3 \cdot 1 = -3$$

Answer: The expression evaluates to -3.

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Solution 5: (j) $\ln 1$

Apply the logarithm of one rule:

$$\ln(1) = 0$$

Answer: The expression evaluates to 0.

Solution 5: (k) $\ln \sqrt{e}$

Rewrite the square root as a power of e, then apply the power rule:

$$\ln\left(\sqrt{e}\right) = \ln\left(e^{\frac{1}{2}}\right) = \frac{1}{2} \cdot \ln(e) = \frac{1}{2} \cdot 1 = \frac{1}{2}$$

Answer: The expression evaluates to $\frac{1}{2}$.

Solution 5: (l) $\ln\left(\frac{1}{e}\right)$

Rewrite the fraction as a power of e, then apply the power rule:

$$\ln\left(\frac{1}{e}\right) = \ln\left(e^{-1}\right) = -1 \cdot \ln(e) = -1 \cdot 1 = -1$$

Answer: The expression evaluates to -1.

Question 6

Find the value of x:

(a) $\log_2 x = 4$

(b) $\log_x 49 = 2$

(c) $\ln(x+1) = 7$

Tip: Converting a Logarithmic Equation to Exponential Form

A logarithmic equation can always be rewritten in exponential form:

$$\log_b(x) = y \quad \Leftrightarrow \quad b^y = x$$

This works whichever of b, x, or y is the unknown: convert first, then solve.

Solution 6: (a) $\log_2(x) = 4$

Convert to exponential form:

$$x = 2^4 \quad \Rightarrow \quad x = 16$$

Answer: $x = 16$.

Solution 6: (b) $\log_x(49) = 2$

Convert to exponential form and solve for x, keeping in mind that a logarithm's base must be positive:

$$49 = x^2 \quad \Rightarrow \quad x = \pm 7 \quad \Rightarrow \quad x = 7$$

Answer: $x = 7$ (the negative root is rejected since a logarithm's base must be positive).

Solution 6: (c) $\ln(x + 1) = 7$

Convert to exponential form:

$$x + 1 = e^7 \quad \Rightarrow \quad x = e^7 - 1 \approx 1095.6332$$

Answer: $x = e^7 - 1 \approx 1095.6332$.

Review 1 - Exponential and Logarithmic Functions

Chapter 2 - Functions

Review 2 - Functions

Chapter 3 - Derivatives

Review 3 - Derivatives

Chapter 4 - Applications of Derivatives

Review 4 - Applications of Derivatives

Exam: 2020 Example

Exam: 2020 January

Exam: 2020 September 1

Exam: 2020 September 2

Exam: 2021 January 1

Exam: 2021 January 2

Exam: 2022 Example

Exam: 2022 January

Exam: 2023 Example

Exam: 2025 January

Exam: 2026 Example

Exam: 2026 January 1

Exam: 2026 August 1

Exam: 2026 August 2

Exam: 2027 Example Midterm

Exam: 2027 Midterm 1

Exam: 2027 Midterm 2

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